

(2017)

Short Questions :

Q2 (a)

- (i) Histogram
- (ii) Formal statistical test
 - * Shapiro-Wilk test
 - * Kolmogorov-Smirnov test

(b) specific statistical functions and techniques you can perform with EDA tools include:

- (i) clustering and dimension reduction techniques.
- (ii) Univariate visualization of each field in raw data set.
- (iii) Bivariate visualization
- (iv) Multivariate visualization
- (v) K-means clustering.
- (v) predictive models.

i) R
ii) python
are two ~~tools~~ main
tools for EDA in
Data analysis.

(c) The ~~Mann~~ Mann-Whitney test is the non-parametric equivalent of the independent samples t-test.

The paired samples Wilcoxon is a non-parametric alternative to paired t-test used to compare paired data.

(d) Nominal and ordinal are non-metric or categorical scales. That is, their response values are not directly usable as a numerical value. Interval and ratio scales are metric scales that allow for various arithmetic operations.

(e)

* Data editor window known as "Data window"

* It is the window you see when you startup SPSS.

* spreadsheet like window is used to enter all the data.

Long Questions

Q3 (a)

(i)

P-value of Kolmogorov-Smirnov = 0.000

p-value of Shapiro-Wilk = 0.000

(ii)

H_0 : Data does not follow the normal distribution.

H_1 : Data follow the normal distribution.

$$\alpha = 0.05$$

$P = 0.000$
We will reject H_0 if $P < 0.05$

As $P < 0.05$

So we reject H_0 .

Hence Data follow the normal distribution.

(iii)

H_0 : Data does not follow the normality.

H_1 : Data follow the normality.

$$\alpha = 0.01$$

$$p = 0.000$$

We will reject H_0 if

$$p < 0.01$$

As $p < 0.01$

So we reject H_0

Hence Data follow the normality at 1% level of significance

Q3(b)

(i)

$$r = 1 - \frac{\delta \Sigma d^2}{n(n^2 - 1)}$$

$$r = -0.191$$

(ii)

$\rho = 0$ (No correlation)

$\rho \neq 0$ (Correlation) (Different @ Correlation)

$$\alpha = 0.05$$

$$t = \frac{r \sqrt{n-2}}{\sqrt{1-r^2}}$$

$$p = 0.596$$

$$r = -0.191$$

We will reject H_0 if

$$p < 0.05$$

As $p > 0.05$

So we do not reject H_0 .

Hence The correlation is zero. (No correlation)

Q4 (a)

4) H_0 : There is no association b/w class rank and smoking habits

H_1 : There is association b/w class ranks and smoking habits.

(ii)

Senior rank students who never smoked

$$\begin{aligned} \% \text{ within smoking habit} &= \frac{47}{283} \times 100 \\ &= 16.6\% \end{aligned}$$

$$\begin{aligned} \% \text{ within class rank} &= \frac{47}{63} \times 100 \\ &= 74.6\% \end{aligned}$$

(iii) How many degree of freedom there

$$\nu = (r-1)(c-1)$$

$$= (4-1)(3-1)$$

$$= 3 \times 2$$

$$= 6$$

(iv)

H_0 : There is no association b/w class rank and smoking habit.

H_1 : There is association b/w class rank and smoking habit.

$$\alpha = 0.05$$

$$\chi^2 = \sum \frac{(f_o - f_e)^2}{f_e} = 307.34$$

$$p = 0.017$$

we will reject H_0 if

$$p < 0.05$$

$$\text{As } p < 0.05$$

So we reject H_0 .

Hence there is association b/w class rank and smoking habit.

$$C = \sqrt{\frac{8\chi^2}{n + \chi^2}}$$

$$0.197 = \sqrt{\frac{\chi^2}{385 + \chi^2}}$$

$$(0.197)^{1/2} = \frac{\chi^2}{385 + \chi^2}$$

$$0.444 = \frac{\chi^2}{385 + \chi^2}$$

$$0.444(385 + \chi^2) = \chi^2$$

$$170.88 + \chi^2 \cdot 0.444 = \chi^2$$

$$\chi^2 \cdot 0.444 - \chi^2 = 170.88$$

$$\chi^2(0.444 - 1) = 170.88$$

$$\chi^2 =$$

$$170.88 = \chi^2 - \chi^2 \cdot 0.444$$

$$= \chi^2(1 - 0.444)$$

$$= 0.556$$

$$\chi^2 = \frac{170.88}{0.556}$$

$$\chi^2 = 307.34$$

Q4 (b)

(i)

H_0 : Two samples are homogeneous.

H_1 : Two samples are not homogeneous.

$$\alpha = 0.05$$

$$\text{Levene Statistic} = 1.708$$

$$p = 0.205$$

We will reject H_0 if

$$p < 0.05$$

$$\text{As } p > 0.05$$

So we Accept H_0 .

And we conclude that two samples are homogeneous.

(ii)

$$H_0: \mu_1 = \mu_2 = \mu_3$$

H_1 : Not all means are equal

$$\alpha = 0.05$$

$$F = \frac{S_1^2}{S_2^2} = 4.414$$

$$p = 0.025$$

We will reject H_0 if $p < 0.05$

$$\text{As } p < 0.05$$

So we reject H_0

Hence There is no significant difference among the exercise groups.

Q5

(i)

Coefficient of determination $r^2 = 0.86$

(ii)

$$\beta = 0.945$$

High school GPA has strong correlation with dependent variable.

$$\beta_{3.12} = 0 \quad (\text{Not significant})$$

$$\beta_{3.12} \neq 0 \quad (\text{significant})$$

$$\alpha = 0.05$$

$$F = \frac{R^2_{3.12} (n-p-1)}{(1-R^2_{3.12})p}$$

$$p = 0.010$$

we will reject H_0 if

$$p < 0.05$$

As $p < 0.05$, we will reject H_0 .
Hence β is significant.

(iii)

$$\hat{y} = a + b_1x_1 + b_2x_2 + b_3x_3$$

$$\hat{y} = -0.411 + 0.002x_1 - 0.002x_2 + 1.201x_3$$

(a) $\beta_1 = 0$ (not significant)
 $\beta \neq 0$ (significant)

$$\alpha = 0.05$$

$$t = \frac{b_1 - \beta_0}{s_b}$$

$$p = 0.4082$$

we will reject H_0 if $p < 0.05$

AS $p > 0.05$

so we accept H_0

Hence Not significant

(b)

 $\beta_2 = 0$ (Not significant) $\beta_2 \neq 0$ (significant)

$$\alpha = 0.05$$

$$t = \frac{\cancel{b_2 - \beta_0}}{S_b} \quad \frac{b_2 - \beta_0}{S_b}$$

$$p = 0.393$$

we will reject H_0 if $p < 0.05$ As $p > 0.05$ so, we Accept H_0 .

(c)

 $\beta_3 = 0$ (not significant) $\beta_3 \neq 0$ (significant)

$$\alpha = 0.05$$

$$t = \frac{b_3 - \beta_0}{S_b}$$

$$p = 0.010$$

we will reject H_0 if $p < 0.05$ As $p < 0.05$ So we reject H_0 .

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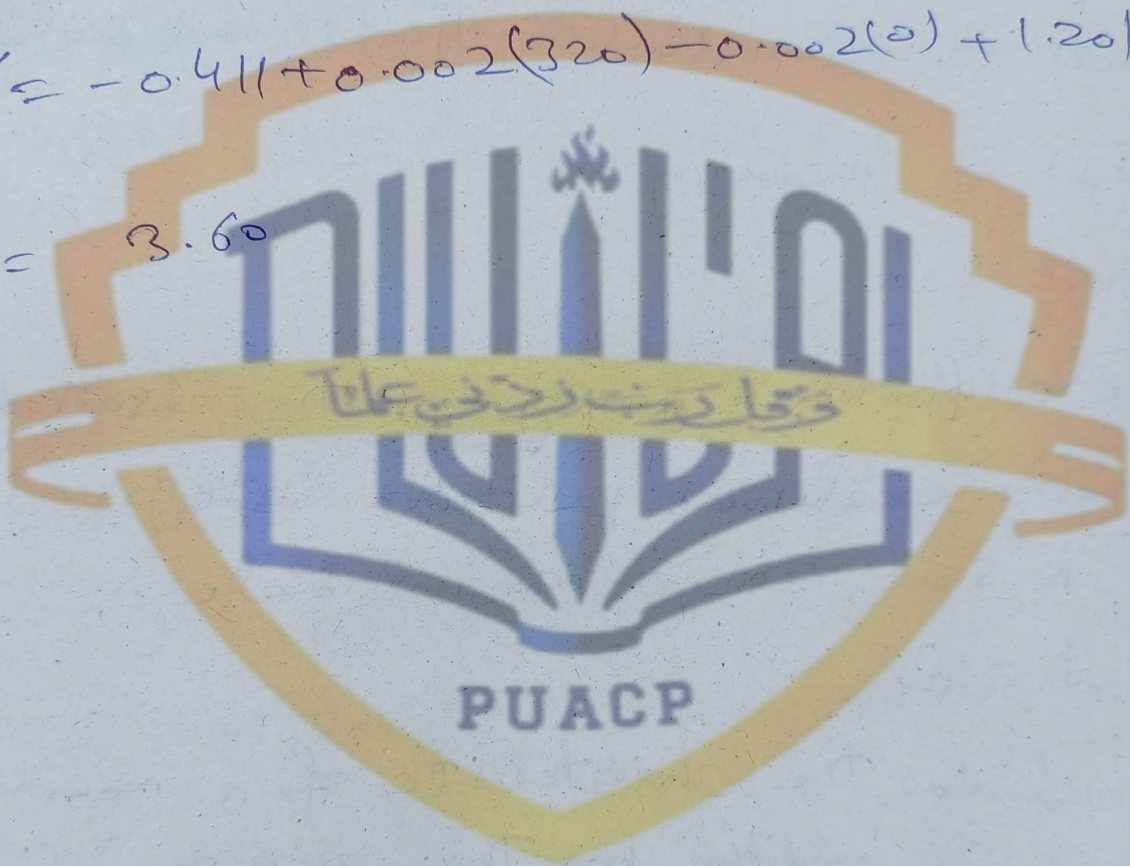
(iv) Put $x_1 = 320$

$$x_2 = 0$$

$$x_3 = 2.81$$

$$\hat{y} = -0.411 + 0.002(320) - 0.002(0) + 1.201(2.81)$$

$$\hat{y} = 3.60$$



(2018)

Q2. i)

By measurement, we usually mean the assigning of numbers to observations or objects and scaling is a process of measuring.

Nominal scale:

The classification or grouping of the observations into mutually exclusive, qualitative categories or classes is said to a nominal scale.

ordinal scale:

It includes the characteristic of a nominal scale and in addition has the property of ordering or ranking of measurements.

Interval scale

A measurement scale possessing a constant interval size but not a true zero point is called an interval scale.

Long Questions

3 (a)

(i)

H_0 : Data does not follow the normal distribution

H_1 : Data follow normal distribution.

$$\alpha = 0.05$$

We use Kolmogorov-Smirnov test to test the normality.

$$P = 0.000$$

We will reject H_0 if

$$P < \alpha$$

$$\text{AS } P < 0.05$$

So we reject H_0

Hence Data follow the normal distribution

(16)

Doc

(ii)

H_0 : The two random samples come from populations having the same distributions

H_1 : The two samples come from populations having different distributions.

$$\alpha = 0.05$$

$$Z = \frac{n_r - \mu_r}{\sigma_r} = -2.158$$

$$p = 0.031$$

We will reject H_0 if

$$p < 0.05$$

As ~~$p > 0.05$~~ $p < 0.05$

So we reject H_0 .

Hence the two samples come from populations having different distributions.

(17)

(iii)

$H = 7.0$ (Not significant)
 $H \neq 7.0$ (Significantly)

$$\alpha = 0.05$$

$$t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}}$$

$$df = n - 1 \\ = 56 - 1 = 55$$

$$P = 0.016$$

We will reject H_0 if

$$P < \alpha$$

$$\text{As } P < 0.05$$

So, we reject H_0 .

Hence Data is significantly different from 7.0

(b)

i)

 H_0 : Two samples are homogeneous. H_1 : Two samples are not homogeneous.

$$\alpha = 0.05$$

$$\text{Levene statistic} = 1.956$$

$$p = 0.154$$

We will reject H_0 if

$$p < \alpha$$

$$\text{As } p > 0.05$$

So we do not reject H_0 .

Hence Two samples are not homogeneous.

(19)

(ii)

$$H_0: \mu_1 = \mu_2 = \mu_3$$

H_1 = Not all means are equal.

$$\alpha = 0.05$$

$$F = \frac{s_1^2}{s_2^2} = 0.217$$

$$p = 0.805$$

We will reject H_0 if

$$p < \alpha$$

As $p > 0.05$

So we do not reject H_0

Hence there is significant difference among the grades of three teachers.

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Q4 (a)

Hobbies	Subjects			
	MATH	Stat	Econo	
Music	5 $\frac{5}{10} \times 100 = 50\%$	2 $\frac{2}{10} \times 100 = 20\%$	3 $\frac{3}{10} \times 100 = 30\%$	10
crafts work	6 $\frac{6}{21} \times 100 = 28.6\%$	8 $\frac{8}{21} \times 100 = 38.1\%$	7 $\frac{7}{21} \times 100 = 33.3\%$	21
Book reading	4 $\frac{4}{15} \times 100 = 26.7$	6 $\frac{6}{15} \times 100 = 40\%$	5 $\frac{5}{15} \times 100 = 33.3$	15
Drama	6 $\frac{6}{14} \times 100 = 42.9$	3 $\frac{3}{14} \times 100 = 21.4$	5 $\frac{5}{14} \times 100 = 35.7$	14
	21	19	20	60

% within hobbies

PUACP

Expected Count Hobbies	Subjects			
	MATH	Stat	Econo	
Music	$\frac{10 \times 21}{60} = 3.5$	3.2	3.3	10
crafts work	7.4	6.7	7.0	21
Book reading	5.3	4.8	5.0	15
Drama	4.9	4.4	4.7	14
	20	19	20	60

% within subjects

Subject

Hobbies	MATH	stat	Econo	
music	5 $\frac{5}{21} \times 100 = 23.8\%$	2 $\frac{2}{19} \times 100 = 10.6\%$	3 $\frac{3}{20} \times 100 = 15\%$	10
crafts works	6 $\frac{6}{21} \times 100 = 28.6\%$	8 $\frac{8}{19} \times 100 = 42.1\%$	7 $\frac{7}{20} \times 100 = 35\%$	21
Book read	4 $\frac{4}{21} \times 100 = 19.0\%$	6 $\frac{6}{19} \times 100 = 31.6\%$	5 $\frac{5}{20} \times 100 = 25\%$	15
Drama	6 $\frac{6}{21} \times 100 = 28.6\%$	3 $\frac{3}{19} \times 100 = 15.8\%$	5 $\frac{5}{20} \times 100 = 25\%$	14
	21	19	20	60

% of total

Subject

Hobbies	MATH	stat	Econo	
music	5 $\frac{5}{60} \times 100 = 8.3\%$	2 $\frac{2}{60} \times 100 = 3.3\%$	3 $\frac{3}{60} \times 100 = 5.0\%$	10
crafts work	6 $\frac{6}{60} \times 100 = 10.0\%$	8 $\frac{8}{60} \times 100 = 13.3\%$	7 $\frac{7}{60} \times 100 = 11.7\%$	21
Book read	4 $\frac{4}{60} \times 100 = 6.7\%$	6 $\frac{6}{60} \times 100 = 10.0\%$	5 $\frac{5}{60} \times 100 = 8.3\%$	15
Drama	6 $\frac{6}{60} \times 100 = 10.0\%$	3 $\frac{3}{60} \times 100 = 5.0\%$	5 $\frac{5}{60} \times 100 = 8.3\%$	14
	21	19	20	60

(22)

(a)

(i)

Percentage of statistics students having crafts work as a hobby from all statistics.

$$\% \text{ within hobbies} = \frac{8}{21} \times 100 = 38.1\%$$

$$\% \text{ within subjects} = \frac{8}{19} \times 100 = 42.1\%$$

(ii)

What percentage of Math students like book reading from all students like book reading

$$\% \text{ within hobbies} = \frac{4}{15} \times 100 = 26.7\%$$

$$\% \text{ within subjects} = \frac{4}{21} \times 100 = 19.0\%$$

(iii) Residual value of economics students with drama.

$$\text{Residual Value} = f_o - f_e$$

$$= 5 - 4.7$$

$$= 0.3$$

(23)

(iv)

H_0 : Hobbies does not depend~~ed~~ upon the Subject

H_1 : Hobbies depend~~ed~~ upon the Subject.

$$\alpha = 0.05$$

$$\chi^2 = \sum \frac{(f_o - f_e)^2}{f_e}, \quad df = (r-1)(c-1)$$

$$p = 0.810$$

we will reject H_0

$$p < 0.05$$

$$\chi^2 = 52.55$$

$$\text{As } p > 0.05$$

So we do not reject H_0 .

hence Hobbies ^{do not} depend upon the subjects.

$$c = \sqrt{\frac{\chi^2}{n + \chi^2}}$$

$$0.218 = \sqrt{\frac{\chi^2}{60 + \chi^2}}$$

$$0.467 = \frac{\chi^2}{60 + \chi^2}$$

$$0.467(60 + \chi^2) = \chi^2$$

$$28.01 + \chi^2 \cdot 0.467 = \chi^2$$

$$28.01 = \chi^2 - \chi^2 \cdot 0.467$$

$$28.01 = \chi^2(1 - 0.467)$$

$$28.01 = \chi^2(0.533)$$

$$\chi^2 = \frac{28.01}{0.533} = 52.55$$

Q4 (b)

- (i) H_0 : Two samples are homogeneous
 H_1 : Two samples are not homogeneous.

$$\alpha = 0.05$$

$$\text{Levene statistic} = 3.246$$

$$p = 0.080$$

we will reject H_0 if

$$p < \alpha$$

$$\text{AS } p > 0.05$$

So we do not reject H_0 .

therefore two blood groups are ~~not~~ homogeneous.

(ii)

 $H_0: \mu_D = 0$ (Not significantly diff)

 $H_1: \mu_D \neq 0$ (Significantly diff)

$$\alpha = 0.05$$

$$t = \frac{\bar{d}}{s_d / \sqrt{n}}$$

$$, \quad \nu = n - 1$$

$$\nu = 20 - 1$$

$$= 19$$

$$t = -0.800$$

$$p = 0.434$$

We will reject H_0 if

$$p < \alpha$$

As $p > 0.05$

So we do not reject H_0 .

Hence there is Not Significantly difference
b/w two blood groups.

(26)

Q.5 (i)

$\rho = 0$ (Not significant)

$\rho \neq 0$ (significant)

$$\alpha = 0.05$$

$$r = 0.906$$

$$t = \frac{0.906 \sqrt{25}}{\sqrt{1 - (0.906)^2}}$$

$$t = \frac{r \sqrt{n-2}}{\sqrt{1-r^2}}, \quad \text{d.f.} = n-2$$
$$= 27-2$$
$$= 25$$

$$t = 10.70$$

$$P = 0.000$$

We will reject H_0 if

$$P < \alpha$$

$$\text{As } P < 0.05$$

So we reject H_0 .

Hence the correlation b/w income and Policies is significant.

(ii) $\rho_{YX_3 \cdot X_1 X_2} = 0$ (Not significant)

$\rho_{YX_3 \cdot X_1 X_2} = 0$ (Significant)

$$t = \frac{0.552 \sqrt{23}}{\sqrt{1 - (0.55)^2}}$$

level of significance

$$t = 3.17$$

2 20.05

$$t = \frac{r_{Y3 \cdot 12} \sqrt{n-2-1}}{\sqrt{1 - r_{Y3 \cdot 12}^2}}, \quad \begin{aligned} n &= n-2-1 \\ &= 27-2-2 \\ &= 23 \end{aligned}$$

PUACP

$$r = 0.552$$

$$p = 0.004$$

We will reject H_0
if $p < 0.05$

$$\text{as } p < 0.05$$

So we reject H_0

Hence Correlation b/w saving and
policies purchases is not significant
when age and income is controlled

(20)

$$\text{vii)} \quad \hat{Y} = a + b_1 x_1 + b_2 x_2 + b_3 x_3$$

$$\hat{Y} = 987.370 - 4.446 x_1 - 1.211 x_2 + 38.664 x_3$$

$$(a) \quad \beta_1 = 0 \quad (\text{Not significant})$$

$$\beta_1 \neq 0 \quad (\text{Significant})$$

$$\alpha = 0.05$$

$$t = \frac{b_1 - \beta_0}{s_b}, \quad df = n - 2$$

$$t = -1.578$$

$$P = 0.128$$

we will reject H_0 if

$$P < \alpha$$

$$\text{As } P > \alpha$$

So we do not reject H_0 .

Hence β_1 is Not significant

(29)

(b) $\beta_2 = 0$ (Not significant)

$\beta_2 \neq 0$ (significant)

$$\alpha = 0.05$$

$$t = \frac{b_2 - \beta_0}{s_b}, \quad \nu = n-2$$

$$t_2 = -0.391$$

$$p = 0.699$$

We will reject H_0 if

$$p < 0.05$$

As $p > 0.05$

So we do not reject H_0

Hence β_2 is not significant

(c)

$\beta_3 = 0$ (Not significant)

$\beta_3 \neq 0$ (significant)

$$\alpha = 0.05$$

$$t = \frac{b_3 - \beta_0}{s_b}$$

$$\nu = n-2$$

$$t = 3.175$$

$$p = 0.004$$

We will reject H_0

if $p < 0.05$

As $p < 0.05$

So we reject H_0

Hence β_3 is significant.

(30)

(iv)

$\beta = 0$ (not significant)

$\beta \neq 0$ (significant)

$$\alpha = 0.05$$

$$F = \frac{\text{Regression MS}}{\text{E MS}}, \quad \nu_1 = 1, \quad \nu_2 = n_2 - 2$$

$$F = 54.499$$

$$P = 0.000$$

We will reject H_0 if

$$P < \alpha$$

$$\text{AS } P \neq 0.000$$

So we reject H_0

hence the regression model is significant.

2019

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Long Questions:

Q3

- (a) H_0 : Data Does not follow the normal distribution
 H_1 : Data follow the normal distribution.

$$\alpha = 0.05$$

we use Kolmogorov-Smirnov

$$p = 0.026$$

we will reject H_0 if

$$p < \alpha$$

$$\text{As } p < 0.05$$

So we reject H_0 .

Hence Data follow the normality.

(b) $\rightarrow$ (Run test)

(32)

H_0 : The two random samples come from populations having the same distribution.

H_1 : The two samples come from populations having different distributions.

$$\alpha = 0.05$$

$$Z = \frac{n_r - H_r}{\sigma_r} = -2.731$$

$$P = 0.006$$

we will reject H_0 if
 $P < \alpha$

$$\text{As } P < 0.05$$

So, we reject H_0 .

Hence, The two samples come from populations having different distributions.

(c)

$$\mu = 200$$

$$\mu \neq 200$$

$$\alpha = 0.05$$

$$t = \frac{\bar{X} - \mu_0}{s/\sqrt{n}} = -3.095$$

$$\begin{aligned} \nu &= n-1 \\ &= 155 \end{aligned}$$

$$p = 0.002$$

We will reject H_0 if

$$p < \alpha$$

$$\text{As } p < 0.05$$

So we reject H_0 .

Then, Avg horsepower of cars is not 200.

Can we say that average horsepower of the cars is 200? (34)

Q4
(ii)



OR

$$r = 0.70$$

strong positive linear relationship.

Strong positive linear relationship

(iii)

$H_0: \rho = 0$ (Not significant)

$H_1: \rho \neq 0$ (Significant)

$$\alpha = 0.05$$

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}, \quad \text{d.f.} = n-2 = 154$$

$$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = \frac{0.70\sqrt{156-2}}{\sqrt{1-0.70^2}} = \frac{0.70\sqrt{154}}{\sqrt{0.51}} = 154$$

calculated $r = 0.710$

If $P < 0.05$ we will reject H_0 .

As $P < 0.05$

So we reject H_0 , it means there are significant relationship.

(a)

35



i)

H_0 : There is ^{no} relationship b/w color and Taste

H_1 : There is ~~is~~ relationship b/w color and taste

(ii) what percentage — — —

~~7.2%~~

$$= \frac{10}{139} \times 100$$

$$= 7.2\%$$

(iii)

Coefficient of Contingency $\Rightarrow C = \sqrt{\frac{\chi^2}{n + \chi^2}}$

$$C = 0.197$$

The Contingency coefficient will be always less than 1 and larger Contingency coefficient the stronger the association. It measures the strength of association.

If C is near zero (or equal to zero) you can conclude that variables are independent and there is no association b/w them.

So we can say that there is relationship b/w color and taste

(36)

Q2

$$P = 0.017$$

$$\text{As } P < 0.05$$

So we reject H_0

and Hence we conclude that there is relationship b/w color and taste

Q4 (b)

(i)

H_0 : Two samples are homogeneous

H_1 : Two samples are not homogeneous

$$\alpha = 0.05$$

~~Levene statistic = 1.708~~

$$P = ~~0.005~~ 0.205$$

$$\text{As } ~~P < 0.05~~ P > 0.05$$

variation

So we ~~reject~~ ^{accept} H_0 .

And we conclude that two samples are ~~not~~ homogeneous.

(ii) one way ANOVA

(2020)

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Long questions:

Q3 (a)

$\beta = 0$ (Not significant)

$\beta \neq 0$ (significant)

$$\alpha = 0.05$$

$$t = \frac{b - \beta_0}{s_b}$$

$$v = n - 2$$

$$b = 0.417$$

~~$t = 0.012$~~

$$p = 0.012$$

We will reject H_0 if

$$p < \alpha$$

As $p < 0.05$

So, we reject H_0 .

Hence independent variable is significant

~~$r^2 = 0.007$~~ ~~$r = 0.84$~~

$$r = 0.084$$

Weak positive relationship of independent variable with dependent variable.

(b)

$$\hat{y} = a + bx$$

$$\hat{y} = 35.589 + 0.417x$$

(c)

Adjusted $r^2 = 0.006$

Standard error = 14.994

If The Adjusted r^2 is high and standard error is reduced then regression model is reliable.

(d) write a short summary of these results

- (i) weak positive linear relationship b/w independent and dependant variable
- (ii) Regression model is less reliable
- (iii) Regression model is significant
- (iv) Independent variable is significant.

14.

$$\cancel{H_0 = H_1 = H_2}$$

H_0 : Physical exercise alleviate depression

H_1 : Physical exercise ^{not} alleviate depression

$$\alpha = 0.05$$

Kruskal wallis test $\Rightarrow H = \frac{(n-1)(S_k^2 - c)}{S_r^2 - c}$

$$\chi^2 = 7.290$$

$$p = 0.026$$

$$\begin{aligned} df &= k-1 \\ &= 3-1 \\ &= 2 \end{aligned}$$

We will reject H_0 if

$$p < 0.05$$

As $p < 0.026$

~~Hence~~

So we Reject H_0 .

Hence, physical exercise not alleviate depression.

(2021)

—————

(40)

Q3.

(a) $\hat{y} = a + bx$

$$\hat{y} = 122.963 + 0.575x$$

put $x = 20$

$$\hat{y} = 122.963$$

$b = 0.575$ (expected change in dependent for unit change in independent variable)

(b)

Standardized slope coefficient (β) = 0.706

(c)

95% Confidence interval for α

$$\alpha \pm t_{\frac{\alpha}{2}, n} S_a$$

$$111.672 < \alpha < 134.255$$

95% Confidence interval for β

$$b \pm t_{\frac{\alpha}{2}, n} S_b$$

$$0.559 < b < 0.591$$

$$r^2 = 0.499$$

It measures the Percentage of variation in the values of dependent variable that can be explained by the variation in the independent variable.

Q4. (b)

Step ① H_0 : Population variances are equal

H_1 : Population variances are not equal

Step ② Level of significance
 $\alpha = 0.05$

Step ③ Test statistic

Covariance statistic (F) = 1.659, $p = 0.203$

Step ④ Decision Rule

We will reject H_0 if

$$p < \alpha$$

Step ⑤ Conclusion.

As $p > 0.05$

So we do not reject H_0 .

Hence, population variances are equal.

(c)

$$H_0 : \mu = 0$$

$$H_1 : \mu \neq 0$$

Step ② Level of significance
 $\alpha = 0.05$

Step ③ Test statistic

$$t = \frac{\bar{X} - \mu}{s/\sqrt{n}}, \quad \nu = n-1$$

$$p = 0.951$$

Step ④ Decision Rule

We will reject H_0 if

$$p < \alpha$$

Step ⑤ Conclusion.

$$\text{As } p > 0.05$$

So, we do not reject H_0 .

2020

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2 (b)

$$H_0: p_1 = p_2$$

$$H_1: p_1 \neq p_2$$

$$\alpha = 0.05$$

$$Z = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\hat{p}_c \hat{q}_c \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

$$\hat{p}_c = \frac{n_1 \hat{p}_1 + n_2 \hat{p}_2}{n_1 + n_2}, \hat{q}_c = \sqrt{1 - \hat{p}_c}$$

We will reject H_0 if

$$|Z| > Z_{\alpha/2}$$

Conclusion.

This test indicates that respondents were different or not different from other group.

(e) ⁽⁴⁴⁾
 H_0 : There is no association b/w two attributes
 H_1 : There is association b/w two attributes.

$$\alpha = 0.05$$

$$\chi^2 = \sum \frac{(f_o - f_e)^2}{f_e}$$

$$\chi^2_{1-\alpha, v} = \chi^2_{tab}$$

We will reject H_0 if

if ~~χ^2~~ $\chi^2_{cal} > \chi^2_{tab}$

Hence, Conclusion

If time is categorized then chi-square is applied to test the association b/w two attributes and their ~~scales~~ scales are interval scales.

(SPSS)

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Non-Parametric

i) Mann-Whitney test:

step ①

H_0 : There is no difference in the distribution of —

OR

There was not significant difference between men and women

H_a : The two distributions OR are different.

There was significant difference b/w men and women

②

level of significance

$$\alpha = 0.05$$

③ Test statistic

$$U_1 = n_1 n_2 + \frac{n_1(n_1+1)}{2} - R_1$$

$$U_2 = n_1 n_2 + \frac{n_2(n_2+1)}{2} - R_2$$

The smaller of two values for U_1 and U_2 is taken as U-statistic.

(46) (5)

~~Mann-Whitney U = ? (value)~~

④ Decision Rule

We will reject H_0 if
 $P < 0.05 (2)$

⑤ calculation.

Mann-Whitney U = Value ?

⑥ Conclusion.

~~A~~ $P = 0.457$

As $P > 0.05$

So we do not reject H_0 .

→ ————— →

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Wilcoxon test : (Paired Observations) (for dependent)

Step ①

H_0 : There is no significant difference b/w _____

$$M = \Delta$$

OR

H_1 : There is significant difference b/w _____

$$M \neq \Delta$$

Step ② Level of significance

$$\alpha = 0.05$$

Step ③ Test statistic

The test-statistic is T , the smaller sum of ranks with the sign ignored.

Step ④ Decision Rule

we will reject H_0 if

$$P < \alpha$$

Step ⑤ Calculation

$Z = \text{value?}$

(From Wilcoxon table)

Step ① Conclusion

$$P = ? \Rightarrow 0.943$$

As $p > 0.05$

So, we do not reject H_0 .

Hence, there is no significant difference b/w

* ————— *

(ii) Kruskal-Wallis test (k-independent samples)

Step ①

$$H_0: \mu_1 = \mu_2 = \mu_3$$

H_1 : At least two of the three are not equal.
↓
category

Step ② level of significance

$$\alpha = 0.05$$

Step ③ test statistic

$$H = \frac{(n-1)(S_k^2 - c)}{S_r^2 - c} \quad \text{where}$$

where $S_k^2 = \frac{\sum R_i^2}{n_i}$, $S_r^2 = \sum r_j^2$

$C = \frac{n(n+1)^2}{4}$, $\chi^2 = \sum \frac{(f_o - f_e)^2}{f_e}$; $df = k-1$

step ④ Decision Rule

we will reject H_0

if $p < \alpha$

step ⑤ calculation

$$\chi^2 = \sum \frac{(f_o - f_e)^2}{f_e}$$

$$\chi^2 = 34.622$$

The calculated value for the Kruskal Wallis assessed for significance using chi-square distribution.

step ⑥ Conclusion

$$p = 0.000$$

As $p < 0.05$

So we reject H_0 .

Hence at least two of the — three are not equal.

✱ ————— ✱

(50)

(iv) Run test:

Step ①

H_0 : The sequence of observations is random

H_1 : The sequence of observation is non-random

Step ②

level of significance

$\alpha =$

Step ③

test statistic

$$Z = \frac{n_r - H_r}{\sigma_r}$$

$$H_r = 1 + \frac{2n_1 n_2}{n_1 + n_2}$$

$$\sigma_r = \sqrt{\frac{2n_1 n_2 (2n_1 n_2 - n_1 - n_2)}{(n_1 + n_2)^2 (n_1 + n_2 - 1)}}$$

Step ④ Decision Rule

We will reject H_0 if

$$p < \alpha$$

Step 4 Decision Rule

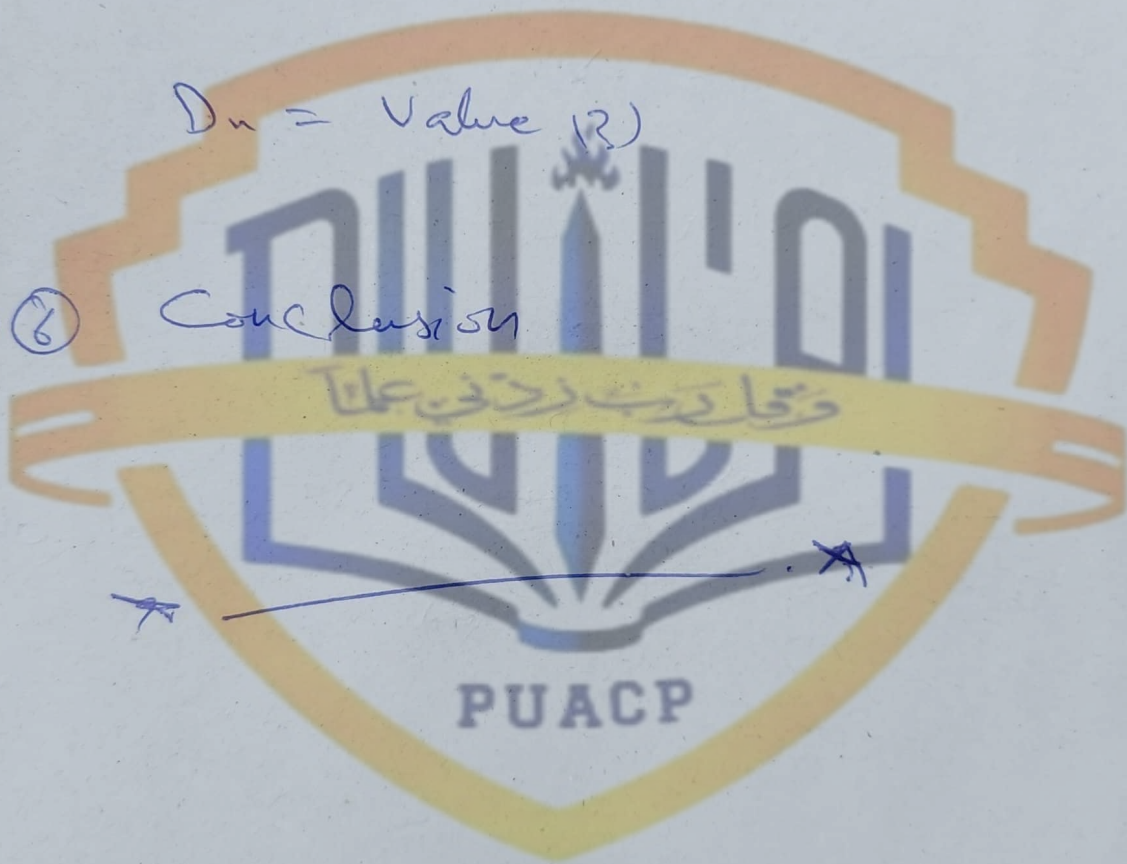
We will reject H_0 if

$$P < 0.05$$

Step 5 Calculation

$$D_n = \text{Value (2)}$$

Step 6 Conclusion



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Step ⑤ Calculation.

$$Z = \text{value } (3)$$

Step ⑥ Conclusion.

(V) Kolmogorov-Smirnov test

Step ①

H_0 : The sample comes from a specified distribution

H_1 : The sample does not come " " " "

Step ② Level of significance.

$$\alpha = 0.05$$

Step ③ Test statistic

$$D_n = \max |S_n(x) - F_0(x)|$$